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CS11021– Programming Laboratory (Intake 43)

Lab Sheet 10

Part I

1. In a data center, servers must operate within a safe temperature range.

Write a C++ program that continuously reads the server temperature in °C and displays the system status.

Conditions:

- **Temperature $< 18^{\circ}\text{C}$ → Display “Warning: Temperature Too Low”**
- **Temperature $18 - 27^{\circ}\text{C}$ → Display “System Operating Normally”**
- **Temperature $> 27^{\circ}\text{C}$ → Display “Alert: Overheating Risk”**

Requirements:

- Use a while loop to continuously monitor temperatures.
- The program should stop when the user enters -1 as the temperature.

2. An Internet Service Provider charges customers based on monthly data usage.

Write a C++ program to calculate the monthly internet bill for multiple customers.

Billing rules:

- **Up to 100 GB → Rs. 500**
- **101 – 300 GB → Rs. 800**
- **301 – 500 GB → Rs. 1200**
- **Above 500 GB → Rs. 1600**

Requirements:

- Input customer name and data usage.
- Display the total bill for each customer.
- Use a do-while loop to repeat the process for multiple customers.
- Ask the user whether they want to continue for another customer (Y/N).

3. A factory machine reports the number of **defective products** detected during production.

Write a C++ program that repeatedly asks the operator to **enter the number of defects per batch**.

Requirements:

- Use a **while loop** to continue entering values until the operator enters **-1**.
- Calculate and display the **total number of defects** and **number of batches inspected**.

4. A smart irrigation system decides whether to **activate water sprinklers**.

Write a C++ program that reads:

- **Soil moisture level (%)**
- **Current temperature (°C)**

Conditions:

- If **moisture < 40 AND temperature > 30** → Display **“Sprinklers Activated”**
- If **moisture ≥ 40** → Display **“Soil Moisture Sufficient”**
- Otherwise → Display **“Monitoring Conditions”**

5. A laboratory uses a **PIN-based door security system**.

Write a C++ program that:

- Allows a user to enter a **4-digit PIN**
- The correct PIN is **2580**
- The user can attempt **maximum 3 times**

Requirements:

- Use a **do-while loop**
- If the correct PIN is entered → Display “**Access Granted**”
- If all attempts fail → Display “**Access Denied**”

6. Drone Battery Monitoring System (While + If-Else Ladder)

A drone continuously reports its **battery level (%)** during flight.

Write a C++ program that:

- Reads the **battery level repeatedly**
- Stops when the battery becomes **0**

Display status based on battery level:

- **> 60%** → “Battery Level: Safe”
- **30% – 60%** → “Battery Level: Moderate”
- **10% – 29%** → “Battery Level: Low – Prepare to Land”
- **< 10%** → “Critical Battery – Immediate Landing Required”

Use a **while loop** for continuous monitoring.

Part II

1. Develop a C++ **function** that simulates a **library entry system** displaying a welcome message when a student scans their ID card at the entrance.
2. Develop a C++ **function** that prints a **cinema ticket summary** after a customer books a movie ticket.
The function should take the **customer name and ticket price** as arguments and display the booking details.
3. Develop a C++ **function** that simulates an **automatic classroom projector** turning on when a lecturer starts a lecture.
4. Develop a C++ **function** that announces a **flight departure at an airport**, displaying the **flight number and departure time**.
5. A delivery company wants to calculate the **total delivery cost** based on the **distance traveled and delivery charge per kilometer**.
Implement this using a **function**.
6. A customer in a **coffee shop** wants to calculate the **final payment amount after adding a service charge percentage**.
Implement this using a **function**.
7. A university needs to compute a **lecturer's final payment** after adding **research allowance and deducting tax**.

Requirements:

Return **Basic Salary** from the **function to the main program**.

Calculate **Final Salary** in **main()**.

8. A weather station needs to **convert rainfall measurement from millimeters to centimeters** and display the result.

Requirements:

Return **Rainfall in millimeters** from the **function to the main program**.

Convert to **centimeters in main()**.

9. A bank wants to compute the **total fixed deposit value after adding annual interest**, then calculate the **final maturity amount**.

Requirements:

Return **Initial Deposit Amount** from the **function to the main program**.

Calculate **Total amount with interest in main()**.

10. A transport company wants to estimate the **cost of operating a delivery vehicle**.

Requirements:

- Return **Distance traveled** from the **function to the main program**.
- In **main()**, calculate:
 - **Fuel consumption**
 - **Fuel cost**
 - **Total travel expense**